Diabetes Prediction using Machine Learning

A Data-Driven Approach for Early Health Assessment

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Project Overview

- **Goal:** Develop a system to identify if a patient is at risk of diabetes.
- **Core Methodology:** Leveraging machine learning for early health assessment.
- **Workflow Steps:**
- Data Loading & Preprocessing.
- Exploratory Data Analysis (EDA).
- Model Training & Evaluation

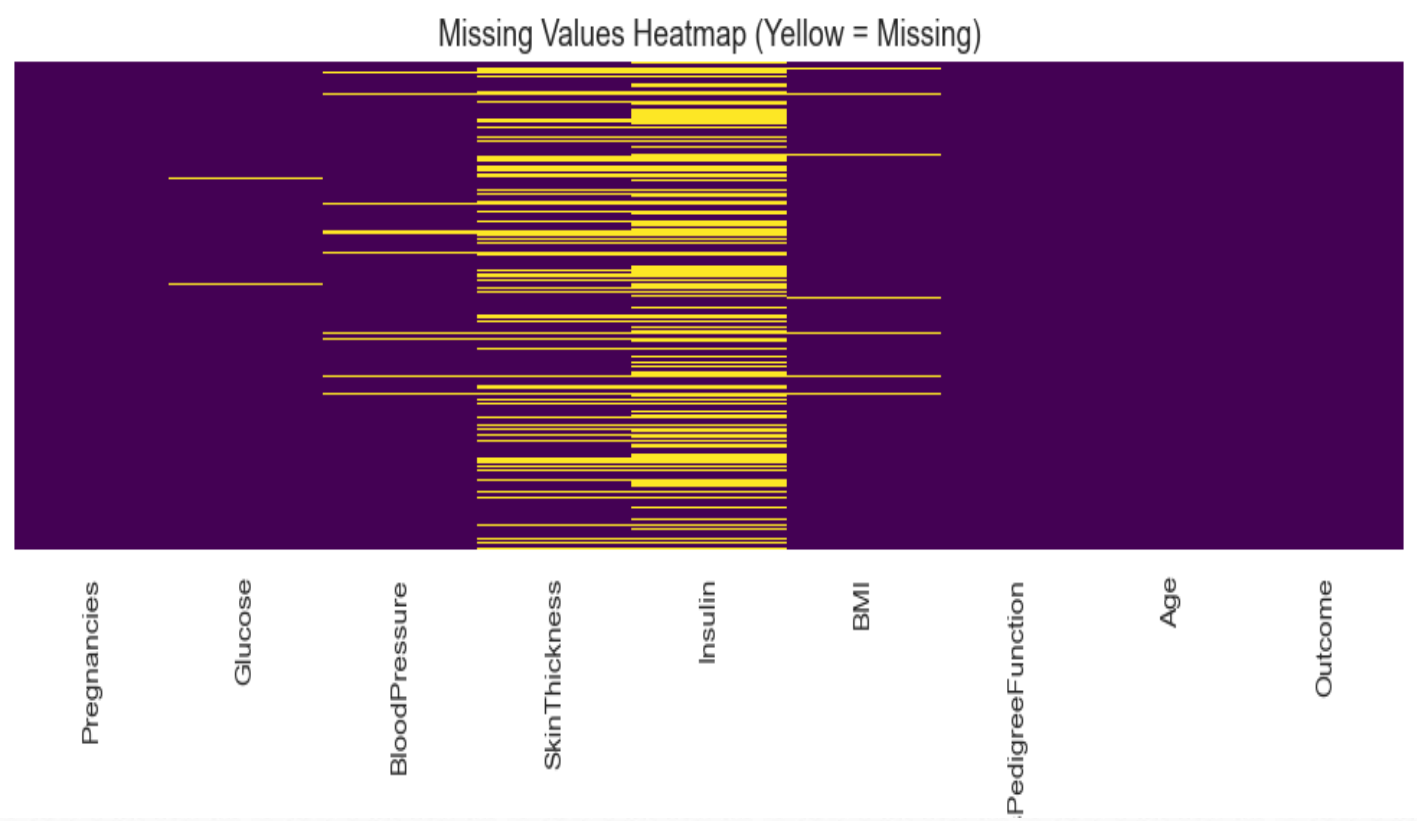
Problem Statement

- **Health Impact:** Diabetes is a chronic disease that can lead to kidney failure, heart disease, and vision loss.
- **The Challenge:** Traditional diagnostic methods are often time-consuming and manual.
- **The Solution:** An automated system to analyze metrics like Glucose and BMI for instant risk assessment.

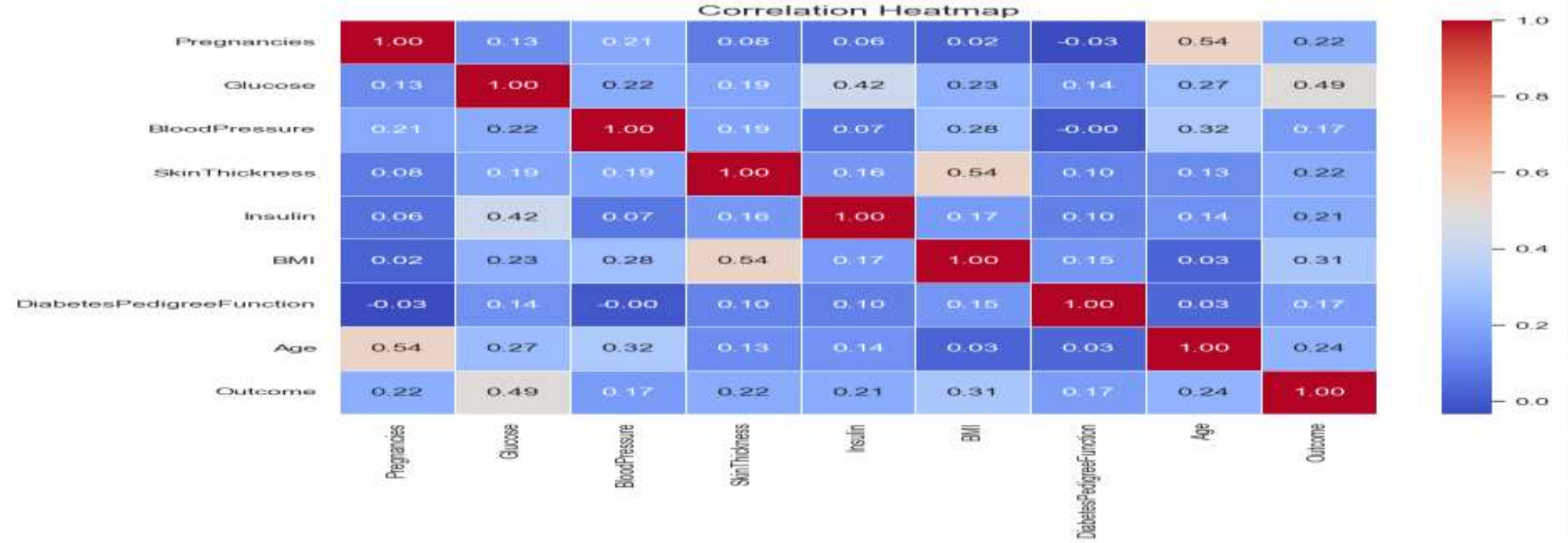
Dataset Source

- **Name:** Pima Indians Diabetes Database.
- **Origin:** National Institute of Diabetes and Digestive and Kidney Diseases.
- **Composition:** 768 clinical observations from female patients.
- **Features:** Includes 8 diagnostic features and 1 binary outcome

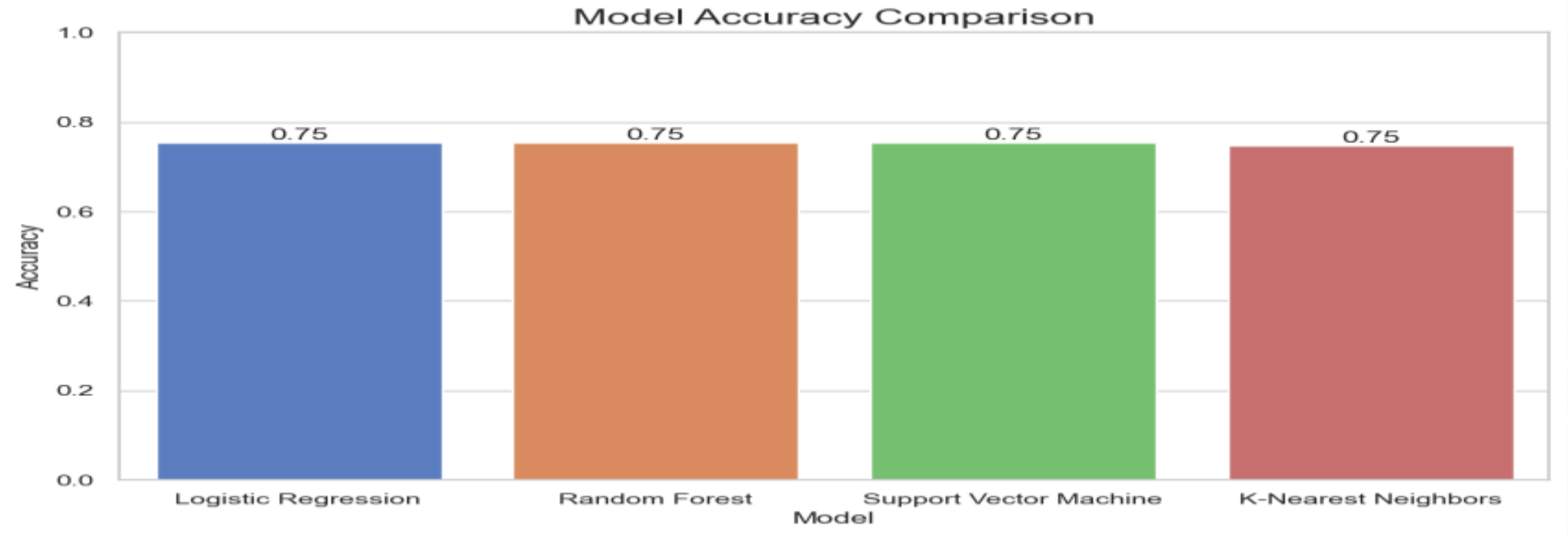
Data Preparation & Preprocessing



Exploratory Data Analysis (EDA)

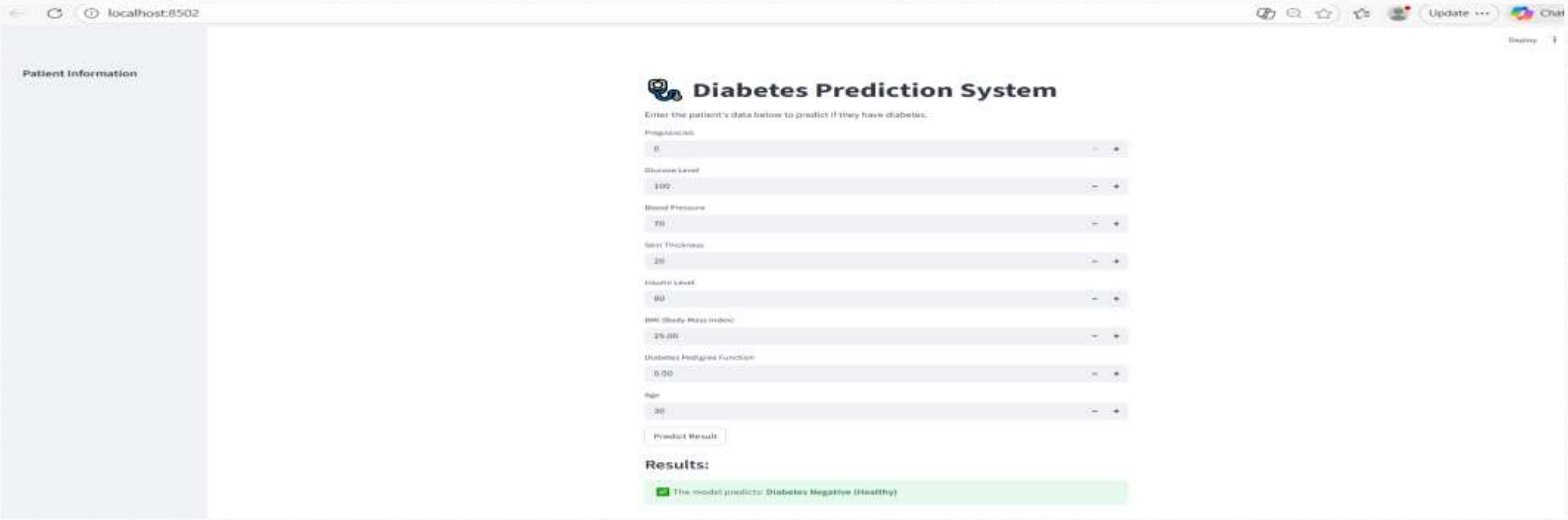


Model Development (Pipeline Approach)



Final Model: Support Vector Machine (SVM)

Deployment (Streamlit Web App)



Conclusion

- **Outcome:** Successfully built an end-to-end workflow with 76% accuracy.
- **Value:** Provides a practical foundation for early diabetes risk detection.
- **Future Impact:** Supports healthcare professionals in prioritizing high-risk cases.

Thanks